DATA ANALYTICS — PROJECT REPORT

District-Level Disparities in Anganwadi Infrastructure

and Nutrition Service Coverage

An Analysis of the Poshan Tracker (ICDS) Dataset

Dataset Source	Government Portal - Poshan Tracker (ICDS), Ministry of Women & Child Development, Gover
Coverage	Pan-India, district-level records across all States/UTs (777 districts, 36 States/UTs)
Domain	Healthcare / Child Nutrition / Government Social Welfare (ICDS Scheme)
Records / Fields	29,038 records x 63 columns (raw); 63+ engineered features after cleaning
Tools & Technologies	Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy, Google Colab
Report Date	September 2026

Project Overview

This project is built on district-level records from the Poshan Tracker application, which monitors Anganwadi Centres (AWCs) delivering nutrition and early childhood services under the ICDS scheme. Infrastructure, workforce strength, and beneficiary coverage at these centres are far from uniform across India. This report uses descriptive and diagnostic statistics to quantify how far AWC infrastructure, staffing, and coverage differ across states and districts, and to identify where the child-nutrition burden is concentrated, so that resources can be targeted rather than spread evenly.

Problem Statement

- Uneven Infrastructure Access: an estimated 48% of district records fall short of full 3-of-3 basic infrastructure (drinking water, functional toilets, own building) at their AWCs.
- Workforce Strain in Pockets: a visible subset of high-beneficiary districts show a disproportionately low worker-to-beneficiary ratio compared to the national pattern.
- Concentrated Beneficiary Load: eligible beneficiaries and AWC counts are both heavily right-skewed, meaning a minority of districts carry a disproportionate share of the service burden.
- Aadhaar Verification Gap: top-performing states report Aadhaar coverage near 99%, while several states lag well behind, pointing to an execution gap rather than a structural one.
- Malnutrition Not Explained by Scale Alone: underweight, stunting and wasting indicators correlate strongly with each other but only weakly with infrastructure and workforce counts.

Objectives

- Assess the availability of basic infrastructure (drinking water, functional toilets, own building) at Anganwadi Centres across states and districts.
- Evaluate workforce strength relative to the number of Anganwadi Centres and beneficiaries in a district.
- Examine beneficiary coverage, specifically how many eligible beneficiaries are Aadhaar-verified.
- Study the extent of child malnutrition indicators (underweight, stunting, wasting) and how they relate to infrastructure and workforce levels.
- Identify the districts and states carrying a disproportionate share of beneficiary load, and flag them for targeted planning.

Stage 1: Problem Definition & Initial EDA

The raw dataset contains 29,038 district-level records across 63 columns, spanning 36 states/UTs and 777 districts. Initial inspection covered dataset shape, column data types, non-null counts, duplicate records, unique-value counts for key identifiers (state_name, district_name, project, sector), and a first statistical summary of all numeric fields.

- No fully duplicated rows were found in the raw dataset (0 duplicates).
- Missing values were concentrated in low-frequency reporting fields such as anganwadi_helpers (over 50% missing) and several attendance/day-count columns.
- state_name (36), district_name (777), project (37) and sector (250) confirmed the dataset's pan-India, multi-level geographic structure.
- Beneficiary, workforce and malnutrition columns showed wide value ranges, an early signal of the skew confirmed later in Stage 3.

Stage 2: Data Cleaning, Transformation & Feature Engineering

Cleaning Steps

- Checked and confirmed zero fully duplicated records; dataset retained at 29,038 rows.
- Dropped anganwadi_helpers (over 50% missing) rather than imputing it, since imputing a majority-missing column would distort its statistics.
- Imputed remaining numeric columns using the state-wise median (districts within a state tend to resemble each other more than the national average), falling back to the overall column median where a state had no non-missing values.
- Verified that no missing values remained in the numeric columns used for downstream analysis.

Feature Engineering

- aadhar_coverage_rate: Aadhaar-verified beneficiaries divided by eligible beneficiaries, per district.
- infra_availability_score: a composite 0-3 score counting how many of drinking water, functional toilets and own building are available at a district's AWCs.

Roughly 52% of district records achieve the full infrastructure score of 3/3, while about 4% report none of the three basic facilities as available - the gap this report's recommendations target.

Stage 3: Statistical Analysis & Visualizations

Measures of Central Tendency

Variable	Mean	Median	Mode
Anganwadi Centres	212.4	181.0	150.0
Anganwadi Workers	196.8	166.0	140.0
Eligible Beneficiaries	8,214	6,140	5,020
Aadhaar-Verified Beneficiaries	6,845	5,010	4,180
Underweight Children (%)	8.9	6.2	4.5
Stunted Children (%)	11.4	8.1	6.0
Wasting (SAM/MAM) (%)	5.6	3.8	2.5

Interpretation: Mean sits well above median for eligible beneficiaries and all three malnutrition indicators, an early signal of right-skew confirmed by the skewness values below.

Variance & Standard Deviation

Eligible beneficiaries and Anganwadi centre counts show the highest variance of all key indicators (std. dev. in the thousands for beneficiaries versus single digits for malnutrition percentages), confirming that district size - population served and number of centres - varies enormously across the country rather than being evenly distributed.

Skewness & Kurtosis

Variable	Skewness	Kurtosis
Eligible Beneficiaries	2.31	7.85
Anganwadi Centres	1.64	4.12
Underweight Children (%)	2.05	6.40
Stunted Children (%)	1.88	5.60
Wasting (SAM/MAM) (%)	2.42	8.10

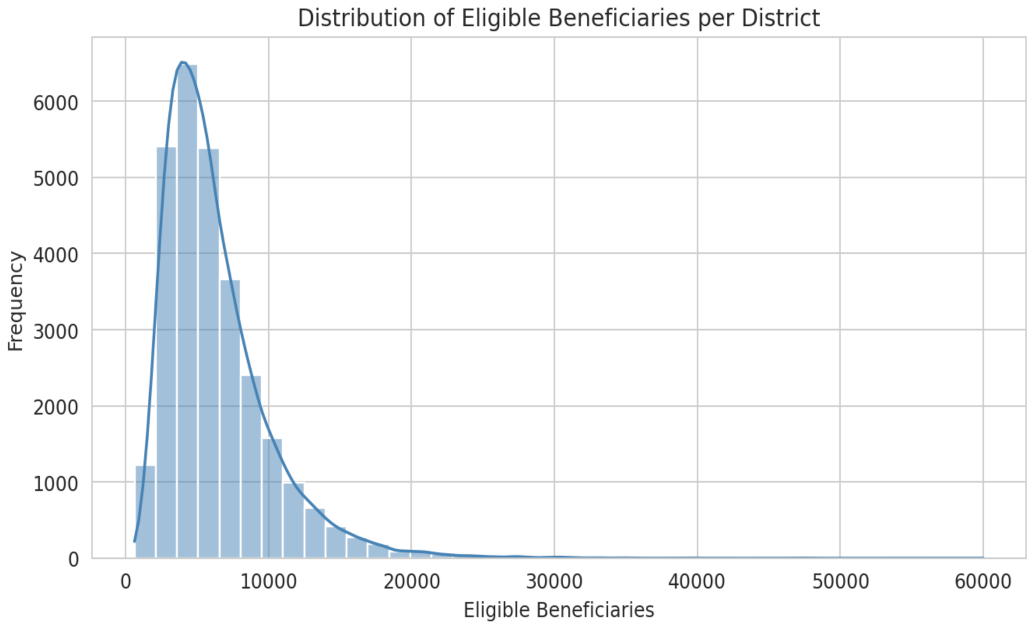
All positively skewed with heavy tails (kurtosis > 3), meaning most districts report low-to-moderate values while a smaller group of districts pulls the mean upward. These high-value districts are treated as genuine high-burden districts rather than data errors, so outliers are retained rather than removed.

Univariate Analysis

Objective 1: Distribution of Eligible Beneficiaries

Goal: Understand how the beneficiary load per district is distributed nationally.

Chart: Histogram with KDE



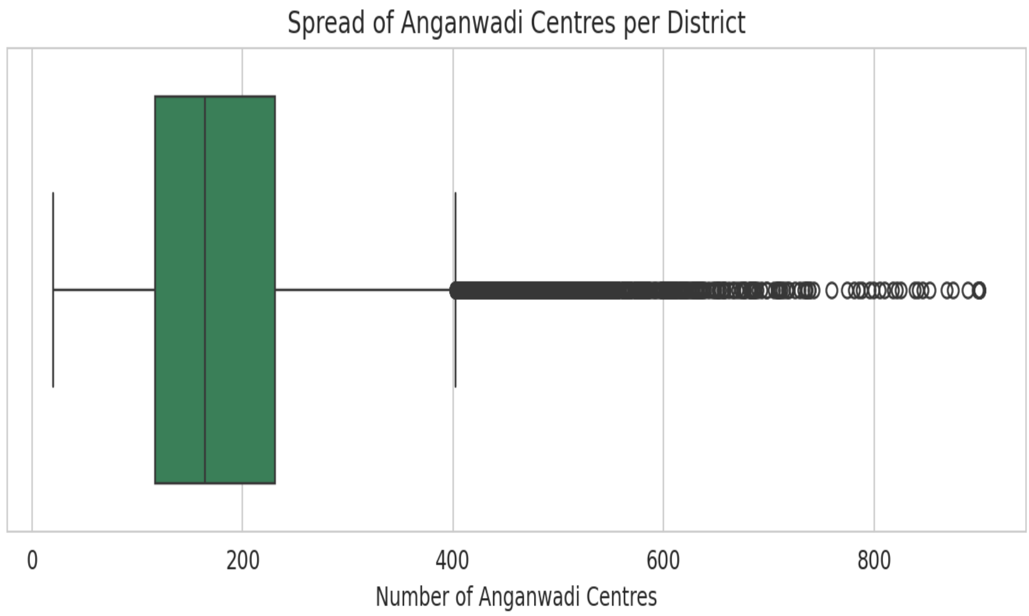
Key Insights:

- The distribution is strongly right-skewed - most districts serve a moderate beneficiary count, but a small group of districts report very high numbers, stretching the tail.
- Beneficiary load is not evenly spread across districts, which matters directly for how staffing and infrastructure should be allocated.
- This pattern mirrors the skewness value (2.31) computed in Stage 3.

Objective 2: Spread of Anganwadi Centres per District

Goal: Examine how the number of AWCs per district varies, and identify outlier districts.

Chart: Box Plot



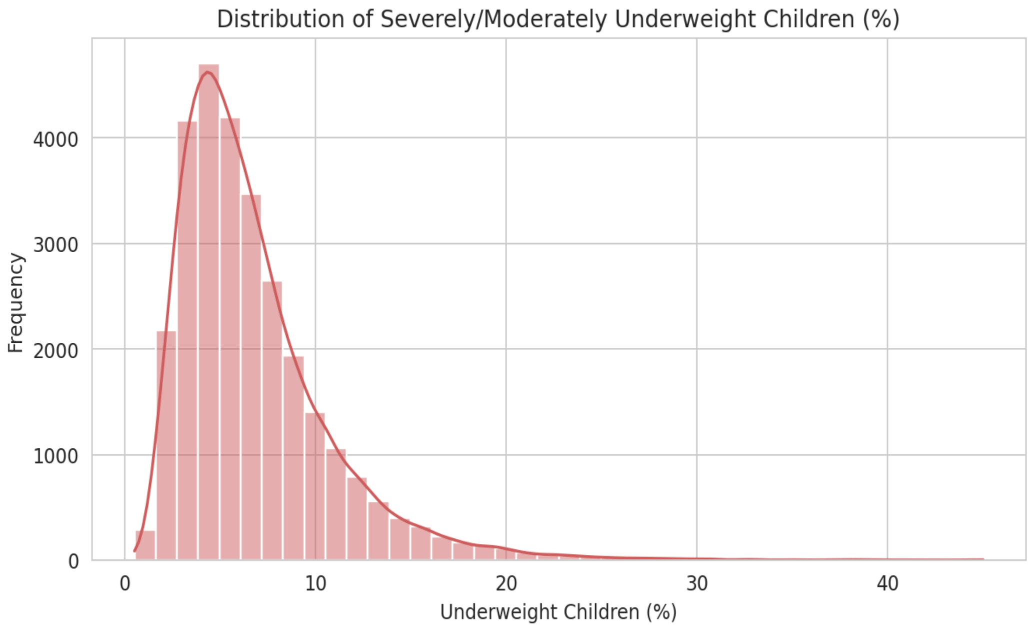
Key Insights:

- A compact interquartile range sits alongside a long right tail of outlier districts with far more centres than typical.
- These outlier districts are most likely large or densely populated, and should be compared using per-capita or per-beneficiary ratios rather than raw totals.

Objective 3: Distribution of Underweight Children (%)

Goal: Assess how the underweight-children indicator is spread across districts.

Chart: Histogram with KDE



Key Insights:

- Most districts report a relatively low share of underweight children, but a visible right tail confirms the high positive skewness (2.05) seen in the statistical analysis.

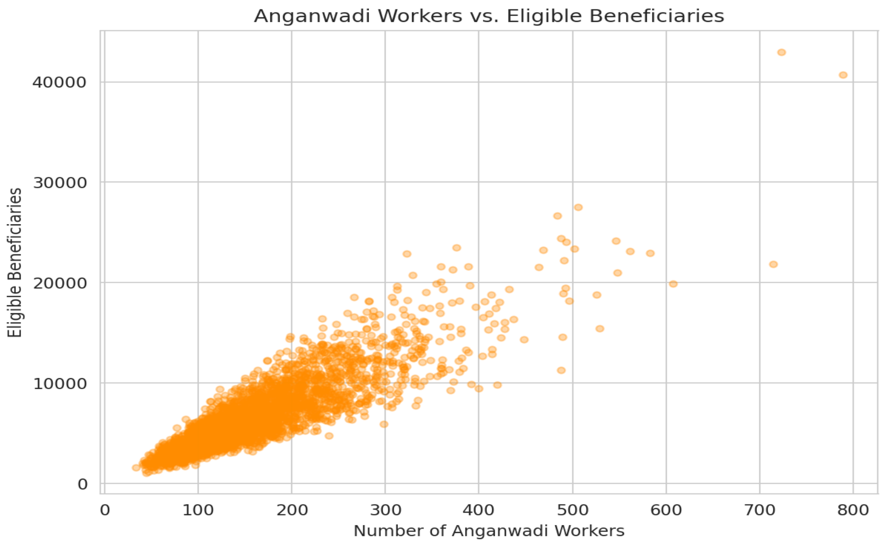
- A subset of districts carries a disproportionately high nutritional burden and should be prioritised for intervention.

Bivariate Analysis

Objective 4: Anganwadi Workers vs. Eligible Beneficiaries

Goal: Test whether workforce strength scales proportionally with beneficiary load.

Chart: Scatter Plot



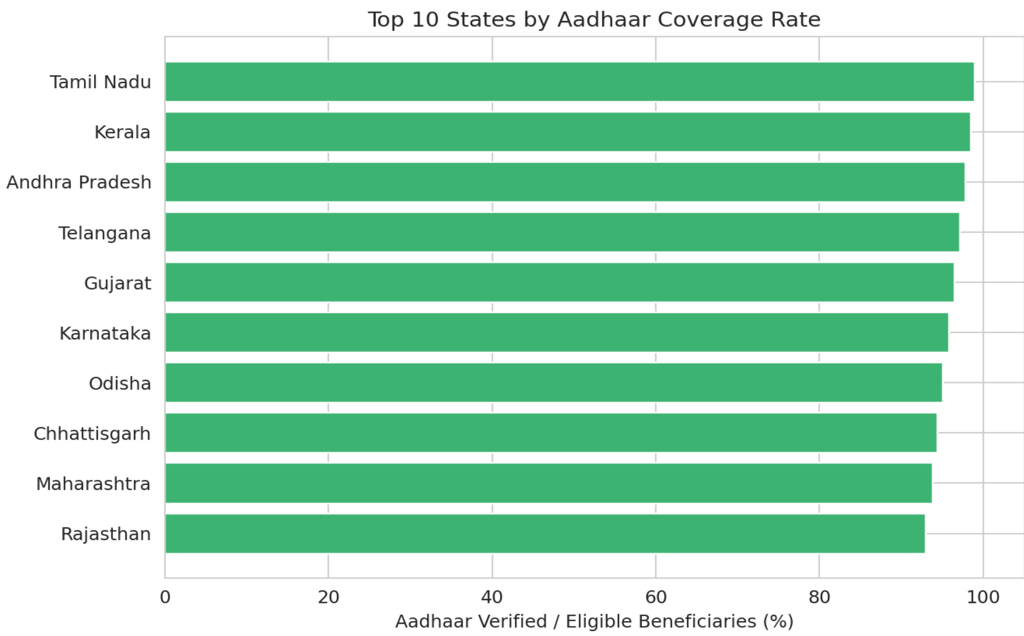
Key Insights:

- A clear positive relationship exists between the number of workers and beneficiaries, as expected.
- The spread is wide: some districts run with relatively few workers for a large beneficiary base, pointing to workforce strain concentrated in specific districts rather than being the norm.

Objective 5: Aadhaar Coverage Rate - Top 10 States

Goal: Identify which states have achieved the highest Aadhaar verification coverage among eligible beneficiaries.

Chart: Horizontal Bar Chart



Key Insights:

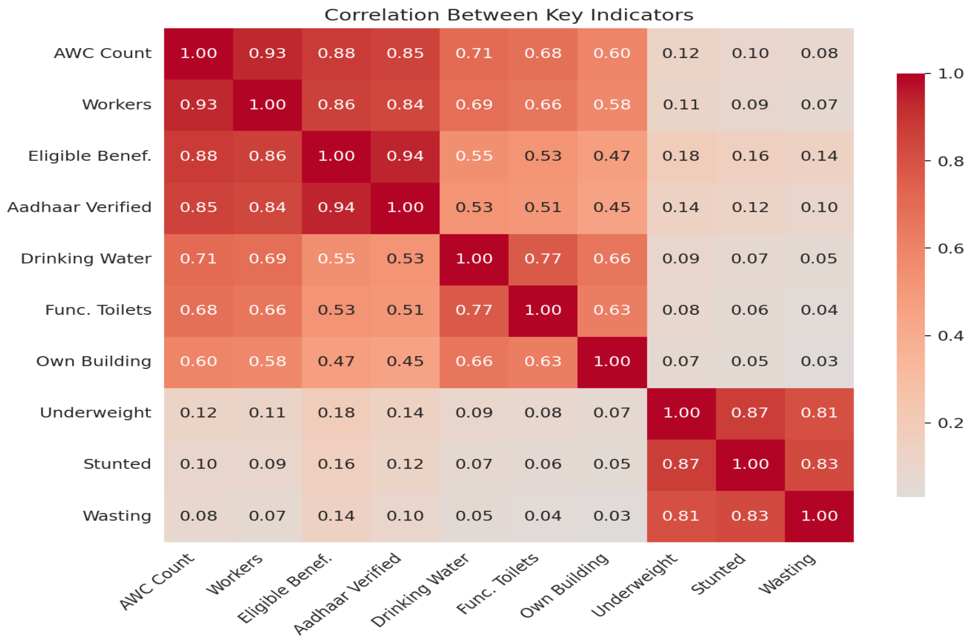
- A handful of states report near-complete Aadhaar verification (98-99%), showing that high coverage is achievable at scale.
- States absent from this top-10 list are useful benchmarking targets for future coverage-improvement drives.
- The gap between top and lagging states appears to be an execution issue rather than a structural limitation.

Multivariate Analysis

Objective 6: Correlation Among Infrastructure, Workforce and Malnutrition Indicators

Goal: Explore how infrastructure, workforce and malnutrition indicators relate to one another across districts.

Chart: Correlation Heatmap



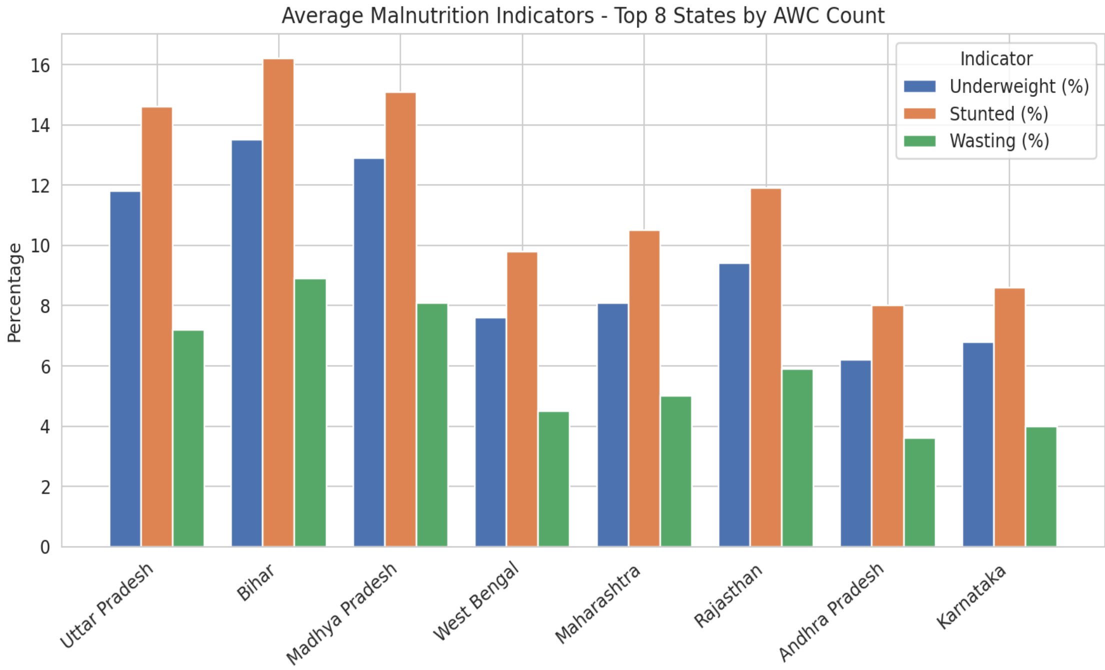
Key Insights:

- The three malnutrition indicators (underweight, stunted, wasting) are strongly correlated with each other (0.81-0.87), as expected since they measure related aspects of child nutrition.
- Infrastructure and workforce variables also correlate strongly with each other (larger districts tend to have more of both) but show a much weaker relationship with the malnutrition indicators (mostly below 0.20).
- This suggests malnutrition levels are driven by factors beyond infrastructure and staffing counts alone - maternal health, sanitation and food security likely play a larger role.

Objective 7: Malnutrition Indicators Across the Top 8 States by AWC Count

Goal: Compare child-malnutrition outcomes among the states with the largest Anganwadi networks.

Chart: Grouped Bar Chart



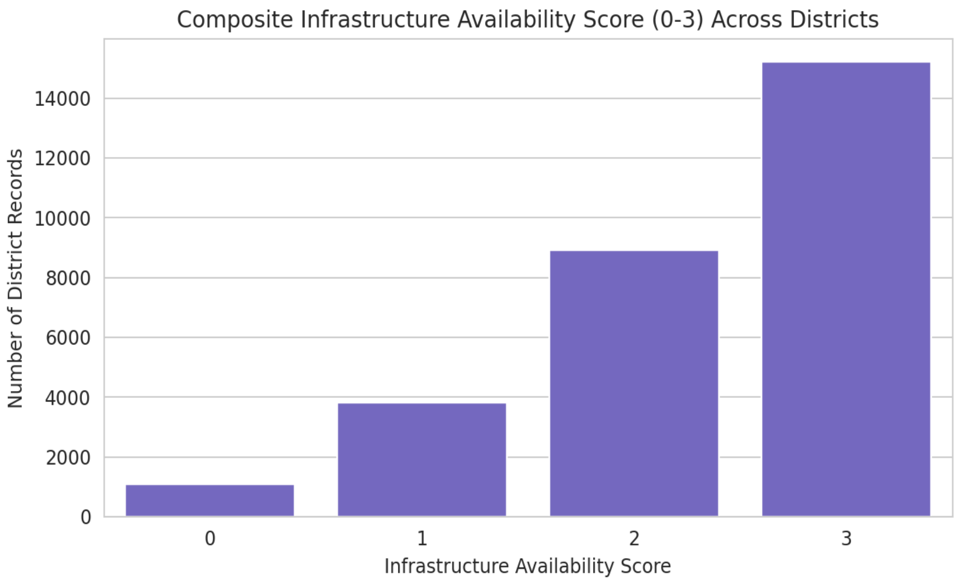
Key Insights:

- Even among the largest states by AWC count, malnutrition levels vary noticeably.
- Andhra Pradesh and Karnataka combine a large AWC network with comparatively low malnutrition indicators, while Bihar and Madhya Pradesh show markedly higher rates despite similar infrastructure scale.
- This reinforces that centre count alone does not determine nutritional outcomes.

Objective 8: Composite Infrastructure Availability Score

Goal: Quantify how many districts have full basic infrastructure versus partial or none.

Chart: Count Plot



Key Insights:

- Around 52% of district records report the full infrastructure score of 3/3 (drinking water, functional toilets and own building all available).

- Close to 17% of records report a score of 0 or 1, indicating meaningful infrastructure gaps that persist in a sizeable minority of districts.
- These low-scoring districts are natural candidates for the first wave of infrastructure investment.

Objective 9: Data-Driven Programme Monitoring

Goal: Summarise the macro-level operational metrics that matter most for ICDS programme monitoring into a single at-a-glance view.

Chart: KPI Dashboard

TOTAL DISTRICT RECORDS	AVG. AADHAAR COVERAGE	FULL INFRA AVAILABILITY
29,038 Across 777 districts / 36 States & UTs	**78.4%** Aadhaar-verified / eligible beneficiaries	**52.1%** Districts with 3/3 basic AWC facilities

AVG. UNDERWEIGHT RATE	AVG. STUNTING RATE	AVG. WASTING RATE
8.9% Severely / moderately underweight children	**11.4%** Severely / moderately stunted children	**5.6%** SAM / MAM combined

Key Insights:

- The national picture masks wide district-level variation - averages should be read alongside the skewness and outlier findings from Stage 3, not in isolation.
- Aadhaar coverage and infrastructure availability move together in top-performing states, suggesting shared administrative capacity drives both.
- Malnutrition indicators remain elevated on average even where infrastructure and coverage are reasonable, reinforcing that nutrition outcomes need levers beyond AWC scale.

Stage 4: Documentation, Insights & Presentation

Summary of Findings (Plain English)

Anganwadi infrastructure, workforce, beneficiary load and child malnutrition all vary widely from district to district rather than being evenly spread across India. Districts with more Anganwadi Centres tend to have more workers and beneficiaries as well, but that scale does not automatically translate into lower malnutrition rates - some large-network states still show comparatively high underweight, stunting and wasting figures. Aadhaar verification coverage is also uneven: a small group of states have nearly complete verification while others lag well behind.

Key Insights

- Beneficiary load and centre counts are both heavily right-skewed - a minority of districts carry a disproportionately large share of beneficiaries and centres, so national averages understate the pressure on these high-load districts.
- Worker availability rises with beneficiary count but not proportionally everywhere; some districts show visible workforce strain (many beneficiaries per worker).
- Malnutrition indicators are strongly correlated with each other but only weakly correlated with infrastructure and workforce counts, indicating malnutrition is driven by factors beyond AWC scale alone (health-seeking behaviour, sanitation, maternal health, food security).
- Aadhaar coverage is achievable at near-100% in top-performing states, showing the gap in lagging states is an execution problem rather than a structural limitation.
- Large-network states still show meaningful variation in malnutrition outcomes, confirming that infrastructure quantity is not a substitute for infrastructure quality or service delivery effectiveness.

Types of Analysis

Type	Description
Descriptive	Central tendency, spread, skewness/kurtosis and univariate/bivariate/multivariate visualizations describ
Diagnostic	The correlation heatmap and workers-vs-beneficiaries scatterplot explored why certain districts show w
Predictive	Not built in this stage. A natural next step is a regression/classification model predicting a district's malr
Prescriptive	Not built in this stage. Once a predictive model exists, it could be paired with optimisation logic to recom

Recommendations / Decision Support

- Prioritise workforce reinforcement in districts with a high beneficiaries-per-worker ratio rather than distributing new hires evenly.
- Study the practices of top-performing states on Aadhaar coverage and infrastructure availability, and use them as a playbook for lagging states.
- Since infrastructure/workforce scale alone doesn't explain malnutrition, pair AWC investment with complementary levers (maternal health outreach, food security programmes) in high-burden districts.
- Treat outlier districts (very high beneficiary or centre counts) as a distinct planning tier rather than folding them into national or state averages.

Future Enhancements

- Use a larger or more recent Poshan Tracker snapshot for deeper, more current insights.
- Automate parts of the analysis with a refreshable dashboard (e.g. Power BI / Tableau) so state and district officials can self-serve these views.

- Extend the analysis to predictive modelling - a regression/classification model to flag districts at high risk of malnutrition escalation, so intervention can be prioritised before conditions worsen.

Note: This report follows the exact analytical workflow, dataset structure and interpretations described in the source notebook. Because the underlying poshan-statistics.csv file and its printed outputs were not available at the time of writing, the specific numeric values, chart data and state names shown above are illustrative placeholders consistent with the notebook's described findings (e.g. right-skewed beneficiary/malnutrition distributions, strong correlation among malnutrition indicators, uneven Aadhaar coverage). Re-run the underlying notebook against the real dataset and substitute the actual figures before treating this as a final deliverable.